

ARCADENEXUS APP

Online Gaming Website
Product Requirement Document (PRD)

Project Title:	ArcadeNexus Online Gaming Website	Document Version:	v1.0.0 (Standard Format)
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Category:	Web Development	Status:	Approved / Verified

1. PROJECT OVERVIEW

ArcadeNexus is a custom online gaming portal designed to run client-side in browsers. It features a responsive layout hosting three interactive games (Tic-Tac-Toe, Snake, and Memory Match) rendered via modals, persisting user scores via the `localStorage` API.

1.1 Who each deliverable is for

Deliverable File	Who it is for
index.html (Semantic Markup) / styles.css	Frontend integration developers and layout designers
app.js & games.js (JS Game Loops)	Developers verifying game controls and local state syncing
assets/ (Static Images & Audio) / README.md	Media designers updating textures; reviewers checking quickstart docs

1.2 Scope shape rationale

Pure client-side vanilla stack (HTML5, CSS3, ES6+ JS) optimized for instant offline load times. No frameworks or external server databases are permitted.

2. DELIVERABLES AT A GLANCE

#	Deliverable	Format	Exact file name
1-2	Structure & Styles	HTML5 / CSS3	`index.html` / `styles.css`
3-4	Control Logic & Games	ES6+ JavaScript	`app.js` / `games.js`
5-6	Docs & Brand Logo	Markdown / SVG	`README.md` / `assets/images/logo.svg`
7-9	Game Cover Cards	PNG Image files	`ttt_card.png` / `snake_card.png` / `memory_card.png` inside `assets/images/`
10-13	Sound Tracks & SFX	MP3 Audio files	`click.mp3` / `score.mp3` / `gameover.mp3` / `cyber_beats.mp3` inside `assets/audio/`

3. BRAND AND LOOK

Deep Void #0B0C10	Cyber Gray #1F2833	Neon Teal #66FCF1	Laser Cyan #45A29E
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Typography: Orbitron for titles/headers/logos; Inter for body copy and statistics tables.

4. THE PRODUCT AND THE STORY

Core facts to show: (1) ArcadeNexus provides a quick-access browser games portal; (2) games operate in overlays; (3) user stats persist locally. Footer line: **Play instantly, challenge your limits on ArcadeNexus.**

4.1 Layout Beats

Beat	Section	What it does & Content Requirements
1-2	Header / Hero	Nav bar with mute control, pulsing neon slogan, CTA link.
3-5	Grid / Dashboard / Footer	3 cards with hover states, stats tables, clear button, licensing notes.

5. DELIVERABLE SPECIFICATIONS

- `index.html`: Fully semantic layout. No inline script variables or onclick event declarations.
- `styles.css`: CSS Variables for palette. Flexible layouts. Media breakpoints: 576px / 992px.
- `app.js` & `games.js`: Class/module encap. Tic-Tac-Toe features active defense/attack AI. Snake features rendering loop and 5% acceleration per 5 foods. Memory Match shuffles cards and enforces 1s delays.

6. WEB QUALITY CHECKS

These checks define standard web layouts and must all pass:

- 1. **Direct DOM rendering:** No external JS frameworks (jQuery, React, Vue) can be imported.
- 2. **Semantic layout:** Uses correct standard structure (header, main, section, footer).
- 3. **Strict separation:** Event listeners must be bound dynamically in JS script files, not in HTML attributes.
- 4. **Zero errors:** Opening index.html directly from directory triggers no runtime or resource errors in Chrome.
- 5. **Responsive fluidity:** Under 576px wide, catalog shifts to single columns without horizontal viewport scrolling.

7. AUDIO SPECIFICATIONS

- **cyber_beats.mp3 (BGM):** Ambient loop sound track set at -20 LUFS. Plays loops automatically when toggled active.
- **UI SFX:** Click SFX (`click.mp3`), score updates (`score.mp3`), game over triggers (`gameover.mp3`) mixed at -14 LUFS.
- **Sound Settings Routing:** Audio elements must route through mute nodes controlled globally via the navbar mute toggle.
- **Licensing:** All assets must be commercial-friendly/Creative Commons, documented in README.md.

8. FILE NAMES AND FOLDER LAYOUT

```
ArcadeNexus/  
  index.html  
  styles.css  
  app.js  
  games.js  
  README.md  
  assets/  
    images/ [logo.svg, tt_t_card.png, snake_card.png, memory_card.png]  
    audio/  [click.mp3, score.mp3, gameover.mp3, cyber_beats.mp3]
```

9. WHAT MUST BE TRUE TO ACCEPT

The delivery is accepted only when all of these are true:

- Unzipped folder is named exactly `ArcadeNexus` and has all files present.
- No off-palette colors show up on visual card surfaces, headers, or text.
- Game modes (TTT, Snake, Memory) load and play smoothly inside overlay modals.
- Stats dashboard persists values using localStorage under the key name `arcadenexus\_stats`.
- Muting sound toggle globally disables background music and click/score/gameover SFX.
- Every rubric item in rubrics.txt passes with verified local execution.